NTIRE 2026 Image Super-Resolution ($\times$4) Challenge Factsheet
PFT-SR with Selective Last-Layer Fine-Tuning

Nihal Kumar
Stony Brook University
nihal.kumar@stonybrook.edu

Snehal Singh Tomar
Stony Brook University
stomar@cs.stonybrook.edu

Klaus Mueller
Stony Brook University
mueller@cs.stonybrook.edu

1. Introduction

This factsheet presents Team VAI-GM's solution for the NTIRE 2026 Image Super-Resolution ($\times$4) challenge. Our final model is built on Progressive Focused Transformer for Single Image Super-Resolution (PFT-SR) [4]. The practical goal of our submission is to achieve strong restoration quality while keeping adaptation and inference reproducible under limited compute and time budgets.

Instead of fully retraining the network, we adopt selective fine-tuning: the early and middle backbone blocks remain frozen, and only the final two transformer blocks with the reconstruction tail are optimized on challenge data. This strategy preserves robust pretrained representations, reduces optimization instability, and gives a favorable quality–efficiency tradeoff for challenge deployment.

2. Factsheet Information

2.1. Team details

- Team name: VAI-GM
- Team ID: 11
- Team leader: Nihal Kumar
- Team leader email: nihal.kumar@stonybrook.edu
- Team members: Snehal Singh Tomar, Klaus Mueller
- Affiliation: Stony Brook University
- NTIRE 2026 Codabench username: nihalk17
- Best scoring entry: 30.99 PSNR on the challenge test set
- Code/executable: https://github.com/imnihalk17/NTIRE2026_ImageSR_x4_submission

2.2. Method details

Overview. Our method uses PFT-SR [4] which derives from BasicSR [5]. We elaborate more on PFT-SR in section 3. The deployed $\times$4 Super Resolution (SR) model configuration uses embedding dimension 240, depth schedule $[4, 4, 4, 6, 6, 6]$, six attention heads, and pixel-shuffle reconstruction. We keep the core architecture unchanged and focus our contribution on a selective fine-tuning strategy that adapts only the final part of the network. We started from pretrained PFT-SR weights, preserved the backbone representation learned during pretraining, and adapted the model on challenge data with a restricted optimization scope. Specifically, we froze earlier and middle backbone stages, then fine-tuned the last two transformer blocks together with the reconstruction tail on Flickr2K. This was followed by reproducible inference and organizer-format packaging for Codabench submission.

Training strategy. We initialize from pretrained PFT-SR weights and fine-tune only the last two transformer blocks plus the SR tail, while freezing earlier blocks. Fine-tuning is performed on Flickr2K for 25,000 iterations. We also tested multiple learning-rate settings during development. In our final setup, we intentionally used a relatively small learning rate because it gave the most stable adaptation for our target use case. In practice, larger learning rates produced faster short-term loss reduction but were less reliable in preserving pretrained knowledge, while the smaller rate reduced catastrophic forgetting during fine-tuning and maintained consistent reconstruction behavior [2]. We chose last-two-block fine-tuning for two main reasons:

- **Richer deep feature composition:** Deeper features already encode stronger composition and aggregation of structures and textures. Adapting these higher-level representations makes fine-tuning more effective than changing early generic feature extractors.
- **Residual correction behavior in late stages:** Final blocks and reconstruction path operate in a residual-learning regime, which aligns with our objective of applying dataset-specific residual corrections during fine-tuning [1].

Overall, this selective adaptation provides a practical balance between quality gain, training stability, and compute efficiency. Our setup summary is as follows:

- Training data: Flickr2K.
- Optimization scope: last 2 transformer blocks + reconstruction tail.
- Frozen scope: early and middle backbone blocks.

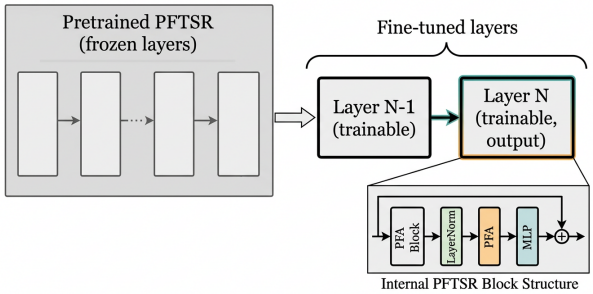

Fig. 1: Proposed PFTSR Fine-tuning Architecture

Figure 1. PFT-SR pipeline used in our selective fine-tuning setup for NTIRE 2026 Image Super-Resolution (×4). The figure borrows the structure of PFT-SR from [4].

- Fine-tuning duration: 25,000 iterations.

Testing description. Inference is single-folder based, mapping an LR input folder to an SR output folder. The inference script auto-detects one checkpoint in `model_zoo/`. If the checkpoint is missing, it reports the download URL from the provided model link file.

Experimental results. During development, our internal validation trajectory reached around 31.25 PSNR. Our final official test-phase score is 30.99 PSNR.

The final challenge outputs are packaged in Codabench format with 100 PNG images and root-level `readme.txt`. The released package corresponds to the final selective fine-tuning run and is generated through our reproducible submission script.

Hardware/testing setup. We performed training and testing iterations in our Colab notebook workflow using NVIDIA A100 GPU instances.

3. Prior Work

PFT-SR [4] forms the bedrock of our approach. It is a Transformer-based super-resolution method designed to recover high-frequency details with progressively focused feature refinement across network depth. At a high level, it combines hierarchical representation learning with attention-driven feature aggregation, allowing the model to recover both local textures and broader structural consistency in a unified framework.

For single-image SR (×4), PFT-SR uses a staged Transformer backbone and a reconstruction head to synthesize high-resolution outputs from low-resolution inputs. Its design emphasizes effective detail restoration while controlling model overhead, making it suitable for practical super-resolution pipelines. This design direction is aligned with modern Transformer-based SR backbones such as SwinIR [3].

We selected PFT-SR for three reasons aligned with our project goals:

- **Strong attention-based restoration behavior:** the model is effective at recovering high-frequency details with structurally consistent outputs.
- **Parameter efficiency in practice:** it provides strong performance without requiring overly large model scaling.
- **Compute-budget friendliness:** it supports practical fine-tuning and inference within typical challenge resource constraints.

These properties make PFT-SR a strong base model for targeted, last-stage adaptation in our final NTIRE 2026 submission.

4. Conclusion and Future Work

In this work, we showed that a selective fine-tuning strategy on top of a strong attention-based SR backbone (PFT-SR) can deliver competitive performance with practical compute constraints. Our final pipeline demonstrates that attention-based SR models can learn dataset-agnostic feature representations that transfer well and can be effectively adapted to related datasets and tasks.

For future work, we plan to scale this direction with stronger compute budgets, improved adapter designs, and larger effective trainable parameter capacity. We expect that fine-tuning PFT-SR with these improvements can further increase reconstruction quality and overall challenge performance.

References

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